

Data Quality Assessment and Improvement Using Tableau Prep (FAIR Dataset)

1. Dataset Description

Title:

Global Terrorism Database (GTD – Sample Extract)

Source:

National Consortium for the Study of Terrorism and Responses to Terrorism

Official Portal: <https://www.start.umd.edu/gtd/>

Dataset Access (CSV): <https://www.kaggle.com/datasets/START-UMD/gtd>

Dataset Overview:

The Global Terrorism Database contains structured records of terrorism-related incidents worldwide. The dataset includes information such as country, year of occurrence, attack type, target type, number of casualties, and weapon type.

Reason for Selection:

This dataset was selected because it is:

- A **FAIR-compliant dataset** (Findable, Accessible, Interoperable, Reusable)
- A real-world dataset used in security and policy research
- Rich in data quality challenges such as missing values, duplicates, and inconsistent formatting
- Suitable for data cleaning using **Tableau Prep**

2. Data Quality Issues Identified

2.1 Missing Values (Completeness)

Several attributes contained missing or incomplete values, particularly:

- Target type
- Weapon type
- Casualty-related fields

These missing values reduce the completeness of the dataset and limit analytical accuracy.

2.2 Inconsistent Formats (Consistency)

- Country names and attack types were not fully standardized
- Variations in categorical labels affected uniform interpretation

2.3 Duplicate Records (Uniqueness)

- Some incidents appeared multiple times due to repeated reporting or data ingestion overlaps
- This affects data integrity and may bias analysis results

2.4 Invalid Values (Validity)

- Casualty values included zero, null, or unrealistic entries
- Some numeric fields were not properly constrained

2.5 Incorrect Data Types (Validity)

- Year field stored as text in some records
- Casualty fields interpreted as strings instead of numeric values

2.6 Irrelevant Fields (Relevance)

- Event identifiers and internal metadata fields were not relevant for analysis objectives
- These fields increased dataset complexity without adding analytical value

3. Data Cleaning Process Using Tableau Prep

The data cleaning workflow was implemented using **Tableau Prep** following structured steps:

Step 1: Data Import

The dataset was imported into Tableau Prep using the Input step. Initial profiling was performed to identify missing values, duplicates, and inconsistent formats.

Step 2: Field Renaming (Traceability Improvement)

Fields were renamed for clarity and better interpretation:

- iyear → Year
- country_txt → Country
- attacktype1_txt → Attack Type
- nkill → Casualties

This improved traceability and readability.

Step 3: Data Type Correction (Validity Improvement)

- Year converted to Integer format
- Casualties converted to Decimal/Number format

This ensured proper analytical processing.

Step 4: Handling Missing Values (Completeness Improvement)

- Records with missing critical fields (Country, Casualties) were filtered or removed
 - This improved dataset completeness and reliability
-

Step 5: Duplicate Removal (Uniqueness Improvement)

- Duplicate records were identified and removed using a composite key (Country + Year + Attack Type)
 - This ensured each incident was uniquely represented
-

Step 6: Filtering Invalid Values (Validity Improvement)

- Records with negative or unrealistic casualty values were removed
 - Extreme outliers were filtered to improve data accuracy
-

Step 7: Removal of Irrelevant Fields (Relevance Improvement)

- Metadata fields such as Event ID and classification codes were removed
 - This reduced dataset complexity and improved focus on analytical variables
-

Step 8: Feature Engineering (Calculated Field)

A new calculated field was created:

Risk Level Classification:

- High Risk → Casualties > 50
- Medium Risk → 10–50
- Low Risk → < 10

This added interpretability to the dataset.

Step 9: Data Export

The cleaned dataset was exported as a CSV file for further analysis and visualization.

4. Data Quality Dimensions Addressed

Dimension	Issue Identified	Improvement Applied
Completeness	Missing values	Filtered or removed NULL entries
Consistency	Inconsistent labels	Standardized field names
Validity	Incorrect formats and values	Type conversion and filtering
Uniqueness	Duplicate records	Removed duplicates
Relevance	Irrelevant metadata fields	Removed unnecessary columns
Traceability	Unclear field names	Renamed variables for clarity

5. Improvements Achieved

The applied data cleaning process resulted in the following improvements:

- Enhanced dataset structure and readability
 - Removal of duplicate and irrelevant records
 - Improved consistency across categorical variables
 - Increased accuracy and reliability of numeric fields
 - Better suitability for visualization and analytical modeling
-

6. Limitations

- Some missing values were removed, resulting in minor data loss
 - No advanced imputation techniques were applied
 - Cleaning was limited to basic Tableau Prep functionalities as per assignment requirements
 - Some inherent inconsistencies in the original dataset remain
-

7. Conclusion

This study demonstrated the application of **Tableau Prep** for improving data quality in a FAIR-compliant real-world dataset. By applying systematic cleaning operations such as filtering, renaming,

type correction, and duplicate removal, the dataset was significantly improved in terms of completeness, consistency, validity, and usability.

The final dataset is clean, structured, and suitable for further analytical processing and visualization, making it a reliable foundation for deriving insights on global terrorism trends.

8. Screenshots

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Tables

- globalterrorismdb_0...

globalterrorismdb_0718dist

136 fields

Filter Values...

3 Recommendations

Data Preview

Field List

Search

Settings

Tables

Data Sample

Changes (1)

Preview

	#	#	#	#
	Source Row Nu...	eventid	year	month
1	1	1,97,00,00,00,001	1,970	7
2	2	1,97,00,00,00,002	1,970	0
3	3	1,97,00,10,00,001	1,970	1
4	4	1,97,00,10,00,002	1,970	1
5	5	1,97,00,10,00,003	1,970	1
6	6	1,97,00,10,10,002	1,970	1
7	7	1,97,00,10,20,001	1,970	1
8	8	1,97,00,10,20,002	1,970	1
9	9	1,97,00,10,20,003	1,970	1
10	10	1,97,00,10,30,001	1,970	1
11	11	1,97,00,10,50,001	1,970	1
12	12	1,97,00,10,60,001	1,970	1
13	13	1,97,00,10,60,001	1,970	1

Connection Details

Text file

File

D:\GERMANY\SRH_MSCADSA\Sem2\DM\100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep\globalterrorismdb_0718dist.csv

Browse

Header Options

☐ Generate header (ex. F1, F2, F3)

☒ Set header

Header and Data Start Row

Header row

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Tables

- globalterrorismdb_0...

Clean 1

135 fields 182K rows

Filter Values...

9 Recommendations

eventid 182K

year 47

imonth 13

iday 32

approxdate 2K

1,97,00,00,00,001

1,970

7

2

null

01/04/2000

01/17/2006

01/28/2002

01/31/2006

02/02/2000

02/07/2002

02/21/2005

02/24/2010

04/11/2007

04/23/2002

04/25/2003

eventid	year	imonth	iday	approxdate	extended	resolution	country	country_txt	region
1 97 00 00 00 001	1 970	7	2	null	0	null	58	Dominican Danvhlir	2

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Tables

- globalterrorismdb_0...

Clean 1 135 fields 182K rows

Filter Values... 17 Recommendations

Search

100%

#	eventid	#	year	#	month	#	day	Abc	approxdate	#	extended
Changes (0)	1,97,00,00,00,001	1,970	7	2	0	0	0	0	0	0	0
	1,97,00,00,00,002	1,970	0	0	0	0	0	0	0	0	0
	1,97,00,10,00,001	1,970	1	0	0	0	0	0	0	0	0
	1,97,00,10,00,002	1,970	1	0	0	0	0	0	0	0	0
	1,97,00,10,00,003	1,970	1	0	0	0	0	0	0	0	0
	1,97,00,10,10,002	1,970	1	1	0	0	0	0	0	0	0
	1,97,00,10,20,001	1,970	1	2	0	0	0	0	0	0	0
	1,97,00,10,20,002	1,970	1	2	0	0	0	0	0	0	0
	1,97,00,10,20,003	1,970	1	2	0	0	0	0	0	0	0
	1,97,00,10,30,001	1,970	1	3	0	0	0	0	0	0	0
	1,97,00,10,50,001	1,970	1	1	0	0	0	0	0	0	0
	1,97,00,10,60,001	1,970	1	6	0	0	0	0	0	0	0
	1,97,00,10,80,001	1,970	1	8	0	0	0	0	0	0	0
	1,97,00,10,90,001	1,970	1	9	0	0	0	0	0	0	0
	1,97,00,10,90,002	1,970	1	9	0	0	0	0	0	0	0

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Tables

- globalterrorismdb_0...

Clean 1 135 fields 182K rows

Filter Values... 17 Recommendations

Search

100%

Abc	approxdate	#	extended	resolution	#	country	Abc	country_txt
Changes (0)	0	0	0	0	58	Dominican Republic		
	0	0	0	0	130	Mexico		
	0	0	0	0	160	Philippines		
	0	0	0	0	78	Greece		
	0	0	0	0	101	Japan		
	0	0	0	0	217	United States		
	0	0	0	0	218	Uruguay		
	0	0	0	0	217	United States		
	0	0	0	0	217	United States		
	0	0	0	0	217	United States		
	0	0	0	0	217	United States		
	0	0	0	0	217	United States		
	0	0	0	0	217	United States		
	0	0	0	0	98	Italy		
	0	0	0	0	217	United States		
	0	0	0	0	217	United States		

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Tables

- globalterrorismdb_0...

globalterrorismdb_0... → Clean 1

Clean 1 135 fields 182K rows Filter Values... 17 Recommendations Search

#	#	#	#	Abc	#
eventid	year	lmonth	iday	approxdate	extended
1,97,00,00,00,001	1,970	7	2	null	0
1,97,00,00,00,002	1,970	0	0	null	0
1,97,00,10,00,001	1,970	1	0	null	0
1,97,00,10,00,002	1,970	1	0	null	0
1,97,00,10,00,003	1,970	1	0	null	0
1,97,00,10,10,002	1,970	1	1	null	0
1,97,00,10,20,001	1,970	1	2	null	0
1,97,00,10,20,002	1,970	1	2	null	0
1,97,00,10,20,003	1,970	1	2	null	0
1,97,00,10,30,001	1,970	1	3	null	0
1,97,00,10,50,001	1,970	1	1	null	0
1,97,00,10,60,001	1,970	1	6	null	0
1,97,00,10,80,001	1,970	1	8	null	0
1,97,00,10,90,001	1,970	1	9	null	0
1,97,00,10,90,002	1,970	1	9	null	0

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Tables

- globalterrorismdb_0...

globalterrorismdb_0... → Clean 1

Clean 1 135 fields 182K rows Filter Values... 17 Recommendations Search

try	Abc	#	Abc	Abc	Abc
country_txt	region	region_txt	provstate	city	
Dominican Republic	2	Central America & Caribbean	null	Santo Domingo	
Mexico	1	North America	Federal	Mexico city	
Philippines	5	Southeast Asia	Tarlac	Unknown	
Greece	8	Western Europe	Attica	Athens	
Japan	4	East Asia	Fukouka	Fukouka	
United States	1	North America	Illinois	Cairo	
Uruguay	3	South America	Montevideo	Montevideo	
United States	1	North America	California	Oakland	
United States	1	North America	Wisconsin	Madison	
United States	1	North America	Wisconsin	Madison	
United States	1	North America	Wisconsin	Baraboo	
United States	1	North America	Colorado	Denver	
Italy	8	Western Europe	Lazio	Rome	
United States	1	North America	Michigan	Detroit	
United States	1	North America	Puerto Rico	Rio Piedras	

[illegible]

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Search

Tables

- globalterrorismdb_0...

globalterrorism... → Clean 1

100%

Clean 1 Showing 1 of 135 fields 182K rows Filter Values... 17 Recommendations Year

Changes (6)

Data Type

- Number (decimal) #
- ✓ Number (whole) #
- Date & Time
- Date
- String Abc

Data Role

- ✓ None
- Geographic
- URL
- Email

1,970
1,970
1,970
1,970

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep

File Edit Flow Server Help

Connections

- globalterrorismdb_0...
Text file

Search

Tables

- globalterrorismdb_0...

globalterrorism... → Clean 1

100%

Clean 1 Showing 1 of 135 fields 182K rows Filter Values... 17 Recommendations cas

Changes (6)

Data Type

- Number (decimal) #
- ✓ Number (whole) #
- Date & Time
- Date
- String Abc

Data Role

- ✓ None
- Geographic
- URL
- Email

0
0
0
0

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0... Text file

Tables

- globalterrorismdb_0...

globalterrorism... Clean 1

100%

Clean 1 135 fields 182K rows Filter Values... 1 Recommendation Search

Filter: Null Values Done

Country 205

Keep Only

Select values to keep.

☐ Null values

☒ Non-null values

country 205

region 12

country Country region region_txt provstate city latitude longitude

58 Dominican Republic 2 Central America & Caribbean null Santo Domingo 19.456702 -60.051164

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorismdb_0... Text file

Tables

- globalterrorismdb_0...

Add Filter

Calculation Must return a boolean value Reference

[casualties] >= 0

ABS

ACOS

AND

AREA

ASC

ASCII

ASIN

ATAN

ATAN2

AVG

BUFFER

CASE

CEILING

CHAR

CONTAINS

COS

COT

COUNT

ABS(number)

Returns the absolute value of the given number.

Example: ABS(-7) = 7

Calculation is valid ^

Create with Tableau Agent

Apply Apply and Close

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

globalterrorismdb_0...
Text file

Search

Tables

globalterrorismdb_0...

globalterrorism... → Clean 1 → Output

100%

Clean 1 135 fields 171K rows Filter Values... 18 Recommendations Search

Changes (15)

- Rename Field
Year
From [year] to [Year]
- Rename Field
Month
From [month] to [Month]
- Rename Field
Day
From [day] to [Day]
- Rename Field
Country
From [country_txt] to [Country]
- Rename Field
AttackType
From [attacktype1_txt] to [AttackType]

Calculation1	Year	Month	Day	app
Low	1,970	7	2	nu
Low	1,970	0	0	nu
Low	1,970	1	0	nu
Low	1,970	1	1	nu
Low	1,970	1	2	nu
Low	1,970	1	2	nu
Low	1,970	1	2	nu
Low	1,970	1	3	nu
Low	1,970	1	1	nu
Low	1,970	1	6	nu
Low	1,970	1	8	nu
Low	1,970	1	9	nu
Low	1,970	1	9	nu
Low	1,970	1	11	nu
Low	1,970	1	12	nu

05:56 PM 30-04-2026

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

globalterrorismdb_0...
Text file

Search

Tables

globalterrorismdb_0...

globalterrorism... → Clean 1 → Output

100%

Clean 1 135 fields 171K rows Filter Values... 18 Recommendations Search

Changes (15)

- Country
From [country_txt] to [Country]
- Rename Field
AttackType
From [attacktype1_txt] to [AttackType]
- Rename Field
Casualties
From [nkil] to [Casualties]
- Change Type
Casualties
To Number (decimal) type
- Filter: Null Values
Country
Keep only: non-null values
- Filter: Null Values
Casualties

Calculation1	Year	Month	Day	app
Low	1,970	7	2	nu
Low	1,970	0	0	nu
Low	1,970	1	0	nu
Low	1,970	1	1	nu
Low	1,970	1	2	nu
Low	1,970	1	2	nu
Low	1,970	1	3	nu
Low	1,970	1	1	nu
Low	1,970	1	6	nu
Low	1,970	1	8	nu
Low	1,970	1	9	nu
Low	1,970	1	9	nu
Low	1,970	1	11	nu
Low	1,970	1	12	nu

05:56 PM 30-04-2026

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

globalterrorismdb_0...
Text file

Search

Tables

globalterrorismdb_0...

globalterrorism... → Clean 1 → Output

100%

Clean 1 135 fields 171K rows Filter Values... 18 Recommendations Search

Changes (15)

- Filter: Null Values
 - Country
 - Keep only: non-null values
 - Casualties
 - Keep only: non-null values
- Rename Field
 - State/Province
 - From [provstate] to [State/Province]
- Filter
 - Casualties
 - [Casualties] >= 0
- Remove Field
 - eventid
- Calculated Field

Calculation1	Year	Month	Day	app
Low	1,970	7	2	nu
Low	1,970	0	0	nu
Low	1,970	1	0	nu
Low	1,970	1	1	nu
Low	1,970	1	2	nu
Low	1,970	1	2	nu
Low	1,970	1	2	nu
Low	1,970	1	3	nu
Low	1,970	1	1	nu
Low	1,970	1	6	nu
Low	1,970	1	8	nu
Low	1,970	1	9	nu
Low	1,970	1	9	nu
Low	1,970	1	11	nu
Low	1,970	1	12	nu

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

globalterrorismdb_0...
Text file

Search

Tables

globalterrorismdb_0...

globalterrorism... → Clean 1 → Output

100%

Clean 1 135 fields 171K rows Filter Values... 18 Recommendations Search

Changes (15)

- [State/Province]
- Filter
 - Casualties
 - [Casualties] >= 0
- Remove Field
 - eventid
- Calculated Field
 - Calculation1
 - IF [Casualties] > 50 THEN "High"
 - ELSEIF [Casualties] > 10 THEN
- Remove Field
 - Calculation1
- Calculated Field
 - Calculation1
 - IF [Casualties] > 50 THEN "High"
 - ELSEIF [Casualties] > 10 THEN

Calculation1	Year	Month	Day	app
Low	1,970	7	2	nu
Low	1,970	0	0	nu
Low	1,970	1	0	nu
Low	1,970	1	1	nu
Low	1,970	1	2	nu
Low	1,970	1	2	nu
Low	1,970	1	2	nu
Low	1,970	1	3	nu
Low	1,970	1	1	nu
Low	1,970	1	6	nu
Low	1,970	1	8	nu
Low	1,970	1	9	nu
Low	1,970	1	9	nu
Low	1,970	1	11	nu
Low	1,970	1	12	nu

Tableau Prep Builder - 100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep*

File Edit Flow Server Help

Connections

- globalterrorisdb_0...
Text file

globalterroris... → Clean 1 → Output

100%

Output 135 fields

Save output to

File

Browse

Name

Output

Location

D:\GERMANY\SRH_MScADSA\SemZ\DM\100005915_Week2-Exercise-Individual-DataQualityCheckwithaFAIRDatasetUsingTableauPrep

Output type

Tableau Data Extract (.hyper)

Write Options

Select an option to create or update your output table.

Run Flow

Save to Output.hyper

Calculation1	Year	Month	Day	approxdate	extended	resolution	country
Low	1,970	7	2	null	0	null	58
Low	1,970	0	0	null	0	null	130
Low	1,970	1	0	null	0	null	160
Low	1,970	1	1	null	0	null	217
Low	1,970	1	2	null	0	null	218
Low	1,970	1	2	null	0	null	217
Low	1,970	1	2	null	0	null	217
Low	1,970	1	3	null	0	null	217
Low	1,970	1	1	null	0	null	217
Low	1,970	1	6	null	0	null	217
Low	1,970	1	8	null	0	null	98
Low	1,970	1	9	null	0	null	217
Low	1,970	1	9	null	0	null	217
Low	1,970	1	11	null	0	null	65